

Problem Statement

AI-Powered Career & Skill-Gap Advisor for Tier-2/3 College Students

Domain:	Education
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Batch:	Batch 2, UPN
Engagement:	AICTE - IBM University Engagement (Edunet Foundation)

The Challenge

Students at tier-2/3 colleges often lack access to personalized career counseling. Placement cells are overloaded, industry skill requirements change rapidly, and students typically rely on scattered, generic advice from forums or seniors. This leads to mismatched career choices and avoidable skill gaps at the time of hiring, disproportionately affecting students without access to structured mentorship.

The Objective

Build an AI-powered Career & Skill-Gap Advisor that delivers personalized, data-backed career guidance to individual students. The solution should:

- **Resume/Skill Analysis** – Parse a student's resume and skill set, and compare it against real, current job-role requirements.
- **Skill-Gap Retrieval (RAG)** – Fetch up-to-date industry skill benchmarks and certification requirements from trusted sources.
- **Learning Path Recommendation** – Suggest specific, relevant courses and certifications to close identified skill gaps.
- **Mock Interview & Feedback** – Simulate role-specific interviews and provide structured, actionable feedback.
- **Multi-Agent System** – Coordinate specialized agents: a Skill Assessment Agent, Career Path Agent, Learning Resource Agent, and Interview Coach Agent.
- **Visualization Dashboard** – Present a skill-gap radar chart, progress tracker, and job-readiness score.

Expected Outcome

An always-available, AI-driven career mentor that reduces dependence on overstretched placement cells, gives every student — regardless of institution tier — access to personalized, evidence-based career guidance, and measurably improves job-readiness before graduation.

Scope Note

This problem statement was independently defined by the student team for the AICTE-IBM University Engagement project submission, in place of a pre-assigned domain problem statement, as permitted under the program's guidelines.